TOPIC 07

Adjusted R-squared

R² that penalises useless variables, and how to compare simple and multivariate regression models fairly

WHAT YOU WILL LEARN

- Explain why R² always rises (or stays the same) when a variable is added
- Write and use the adjusted R² formula
- Show why adjusted R² ≤ R² and when it can fall
- Interpret the lecture's SAT + random-variable example
- Choose between a simple and a multivariate model
- Know the limits of adjusted R²

NOTE **Where this connects**

Topic 04 introduced R² (explained variability, SSR/SST) and Topic 06 added more independent variables. Combining them creates a problem: R² rewards you for *every* variable you add, useful or not. The adjusted R² fixes that, and it is the tool for comparing models, one of the stated aims of this regression course.

1. Two facts about R²

1. R² measures **how much of the total variability is explained by our model**.
2. Adding a variable can **only increase or leave R² the same**. A multivariate regression therefore never has a lower R² than the simple model it extends.

The second fact is a trap. It makes it look as if multivariate models are 'always better', which is false. A variable with no real predictive power still gets a small non-zero coefficient by chance, and R² creeps up. We need a measure that **penalises excessive use of variables**: the adjusted R².

2. The adjusted R²

$$\bar{R}^2 = 1 - (1 - R^2)\frac{n-1}{n-k-1}$$

Here n is the number of observations and k the number of independent variables. Equivalent form: R̄² = 1 − (SSE/(n − k − 1)) / (SST/(n − 1)). In words, R² compares raw sums of squares, while the adjusted version compares **variances** (each divided by its degrees of freedom).

The intuition: each extra variable costs one degree of freedom (Topic 05), so it must improve the fit by more than the cost before the adjusted R² rises.

- **R̄² ≤ R²** always (for k ≥ 1), because (n − 1)/(n − k − 1) is greater than 1.
- The penalty grows with k and shrinks as n grows. With very large n, R̄² ≈ R².
- R̄² **rises** when a variable is added only if it improves the fit by more than chance would; otherwise it falls.

- It can be **negative** when R^2 is small relative to $k/(n-1)$, that is, when the variables add less than chance alone would.

R^2	n	k	Adjusted R^2
0.6	50	1	0.5917
0.6	50	4	0.5644
0.6	50	10	0.4974
0.6	12	4	0.3714
0.05	10	4	−0.7100

The same $R^2 = 0.60$ becomes less impressive when it was achieved with more variables or fewer observations. The last row shows a negative value.

3. The lecture example: GPA = f(SAT) and GPA = f(SAT, Rand)

To see the penalty at work, the lecturer adds a variable that **cannot** help: **Rand 1,2,3**, a number from 1 to 3 assigned **randomly** to every student. We are certain it has no relation to GPA. Compare the two regressions in the lecture workbook.

	Model 1: $GPA = b_0 + b_1SAT$	Model 2: $GPA = b_0 + b_1SAT + b_2Rand$
R^2	0.406	0.407
Adjusted R^2	0.399	0.393
Coefficient of SAT (t, p)	$t = 7.45$, $p \approx 0$	$t = 7.46$, $p \approx 0$
Coefficient of Rand (t, p)	—	$t = 0.42$, $p = 0.677$

- **R^2 went up** slightly (0.406 to 0.407), as it must.
- **Adjusted R^2 went down** (0.399 to 0.393): 'we were penalised for adding a variable that had no explanatory power'.
- The coefficient of Rand has $p = 0.68$, so we cannot reject H_0: $\beta_2 = 0$ (Topic 05). The test agrees with the adjusted R^2: the variable is useless, so **exclude it**.

The lecture's conclusion: always exclude useless information, and use the adjusted R^2 whenever you compare models with different numbers of variables, above all in multivariate regression.

> **NOTE** **A check of the workbook numbers**
> The workbook says 100 observations for Model 2, but its adjusted R^2 of 0.39265 is reproduced exactly only with $n = 84$: $1 - (1 - 0.407285) \times 83/81 = 0.39265$. With $n = 100$ the formula gives 0.3951. The SAT and GPA data in the workbook have 84 rows, so 84 is the real sample size.

4. Our own experiment with random variables

One random variable proves little, so we repeat the lecture's experiment 2,000 times on the 84 students: each time add a fresh variable of random 1s, 2s and 3s, refit, and record how R^2 and adjusted R^2 change relative to the SAT-only model ($R^2 = 0.4060$, adjusted = 0.3988).

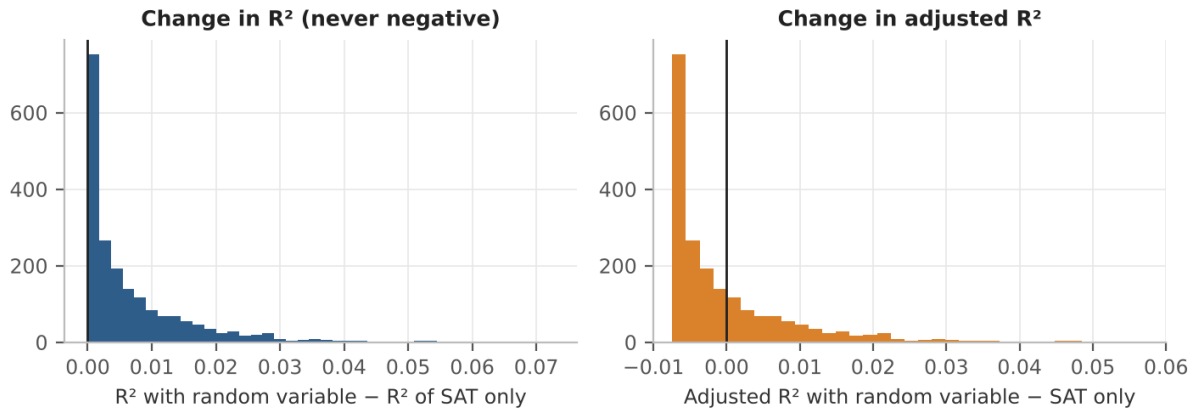


Figure 1. 2,000 pure-noise variables added to the SAT model. R^2 never falls. Adjusted R^2 falls (by a small amount) in most runs but rises by chance in about a third of them.

Result over 2,000 random variables	Value
Times R^2 fell	0
Average increase in R^2	0.0070
Share of runs where adjusted R^2 fell	67.45%
Share of runs where adjusted R^2 rose	32.55%

KEY IDEA **A useful rule**

Adding a variable raises adjusted R^2 **if and only if the absolute value of its t-statistic exceeds 1** (the t-statistic of Topic 05). We checked this in all 2,000 runs. For pure noise $|t|$ exceeds 1 about 32% of the time, matching the share above. So adjusted R^2 is a mild penalty, **not** a significance test: a noise variable can still raise it. Use it together with the coefficient's p-value and with subject knowledge.

5. How to use the adjusted R^2

Situation	Use
Compare models for the **same y** and the **same observations (rows)** with different numbers of variables	Adjusted R^2: prefer the higher one, other things equal
Report how much variability a multivariate model explains	Adjusted R^2
Simple regression only	R^2 and adjusted R^2 are almost the same; either is fine
Judging whether a model is correct or causal	Neither; use theory, residual checks and design

- Do **not** compare R^2 or adjusted R^2 across models with different dependent variables (for example y versus log y).
- Higher adjusted R^2 is not a licence to keep a variable with an insignificant coefficient and no reason to be there.
- Other tools for model choice exist: AIC, BIC and, in machine learning, cross-validation on held-out data.

6. Simple versus multivariate: the choice

1. Start from theory: which variables plausibly affect y?
2. Fit the simple and the multivariate model.
3. Compare **adjusted R^2**; check each new coefficient's p-value and sign.
4. Keep a variable only if it earns its place; otherwise drop it.
5. Prefer the simpler model when the fit is about equal.

7. In Python

```
import numpy as np
import statsmodels.api as sm

data = np.loadtxt("satgpa.csv", delimiter=",", skiprows=1)
sat, gpa = data[:, 0], data[:, 1]
rand = np.random.default_rng(0).integers(1, 4, len(sat))   # random 1, 2, 3

m1 = sm.OLS(gpa, sm.add_constant(sat)).fit()
m2 = sm.OLS(gpa, sm.add_constant(np.column_stack([sat, rand]))).fit()
print(m1.rsquared, m1.rsquared_adj)
print(m2.rsquared, m2.rsquared_adj)        # R2 up, adjusted usually down
print(m2.tvalues, m2.pvalues)             # the random variable's t and p

# by hand
n_, k = len(gpa), 2
print(1 - (1 - m2.rsquared) * (n_ - 1) / (n_ - k - 1))
```

SUMMARY

- R^2 can only rise or stay equal when variables are added, so it cannot compare models with different numbers of variables.
- Adjusted $\bar{R}^2 = 1 - (1 - R^2)(n - 1)/(n - k - 1)$ penalises extra variables; $\bar{R}^2 \leq R^2$.
- Lecture example: adding a random variable raised R^2 (0.406 to 0.407) but lowered adjusted R^2 (0.399 to 0.393).
- Adjusted R^2 rises exactly when the new variable's $|t| > 1$; it is a mild penalty, not a significance test.
- Exclude useless variables, and use adjusted R^2 with p-values and theory when comparing models.

Practice questions

Try each question before reading the answer underneath it.

Q1. $R^2 = 0.60$, $n = 50$, $k = 4$. Find the adjusted R^2.

Answer. $1 - 0.40 \times 49/45 = 0.5644$.

Q2. Model A has $R^2 = 0.52$ with 2 variables, model B has $R^2 = 0.53$ with 8 variables ($n = 30$). Which has the higher adjusted R^2?

Answer. A: $1 - 0.48 \times 29/27 = 0.4844$. B: $1 - 0.47 \times 29/21 = 0.3510$. Model A is better by this measure.

Q3. Why can adjusted R^2 be lower than R^2 but never higher (for $k \geq 1$)?

Answer. The multiplier $(n - 1)/(n - k - 1)$ is greater than 1, so $1 - (1 - R^2) \times$ multiplier is smaller than R^2.

Q4. A new variable has $t = 0.6$. Will adjusted R^2 rise or fall?

Answer. Fall: it rises only when $|t| > 1$.

Q5. A colleague says 'my model has $R^2 = 0.9$ so it is perfect'. Give two reasons to doubt this.

Answer. R^2 rises with every variable even useless ones, so check adjusted R^2; and a high R^2 says nothing about causation, overfitting or whether the model's conditions hold.